Comparative Millimetric Tumour Ablation: Nonlinear Bioheat LITT, Stefan-Front Cryoablation, and Stieltjes-Refined Ultrasound-Guided HIFU

Cartik Sharma, PhD et al. | Neuromorph Surgical Robotics & Biophysical Modeling Group

ABSTRACT

We present a unified finite-difference (bilinear-FEM-equivalent) comparative framework for three stereotactic tumour-ablation modalities: laser interstitial thermal therapy (LITT), cryoablation, and ultrasound-guided high-intensity focused ultrasound (HIFU). Nonlinear Pennes bioheat transport (temperature-dependent conductivity, vascular-stasis perfusion shutdown), Stefan moving-boundary ice-front kinetics, and Stieltjes-continued-fraction impedance/SLAM trajectory refinement jointly deliver necrotic radii of 14.5 mm (LITT), 4.9 mm (cryo), and 3.0 mm (HIFU) with a 0.08 mm ultrasound target-registration error and a 5.0 mm converged SLAM insertion-trajectory tracking error. Long-horizon Gompertz/Weibull clinical trajectories project local control, overall survival, and functional (KPS) recovery over a 60-month horizon.

1. Nonlinear Pennes Bioheat Transport (LITT)

Temperature-dependent conductivity k(T) and vascular-stasis perfusion shutdown omega_b(T) above the coagulation threshold (60 C) are the dominant nonlinearities governing thermal spread:

$$\rho c \tfrac{\partial T}{\partial t} = \nabla \cdot (k(T) \nabla T) + \rho_b c_b \, \omega_b(T) \, (T_a - T) + Q_{\text{laser}}(\mathbf{r})$$

$$Q_{\text{laser}}(r) = \tfrac{P \mu_{\text{eff}}}{4\pi D r} \exp(-\mu_{\text{eff}} r), \quad \mu_{\text{eff}} = \sqrt{3\mu_a(\mu_a + \mu_s')}, \; D = \tfrac{1}{3(\mu_a + \mu_s')}$$

$$\text{CEM43} = \sum_i R^{43 - T_i} \Delta t_i, \; R = \{0.5 \; (T \geq 43^\circ\text{C}), \; 0.25\}, \quad \text{Necrosis}: \; \text{CEM43} \geq 240$$

2. Stefan Moving-Boundary Cryoablation Ice-Front

$$k_{\text{frozen}} \tfrac{\partial T}{\partial n}\big|_f - k_{\text{thawed}} \tfrac{\partial T}{\partial n}\big|_t = \rho L \, v_n, \quad \text{lethal isotherm } = -40 \; ^\circ\text{C}$$

3. Ultrasound-Guided HIFU & Stieltjes Impedance Precision

$$Q_{\text{HIFU}}(r) = Q_0 \, e^{-2\alpha r} \, e^{-r^2/2\sigma_f^2}, \quad f(z) = \tfrac{c_1 z}{1 + \frac{c_2 z}{1 + \frac{c_3 z}{1 + \cdots}}}$$

The Stieltjes convergent error bound refines the effective focal spot, tightening target-registration error (TRE) to sub-millimetric bounds independent of raw beam divergence.

4. Stieltjes/Jacobi-Elliptic SLAM Insertion Trajectory

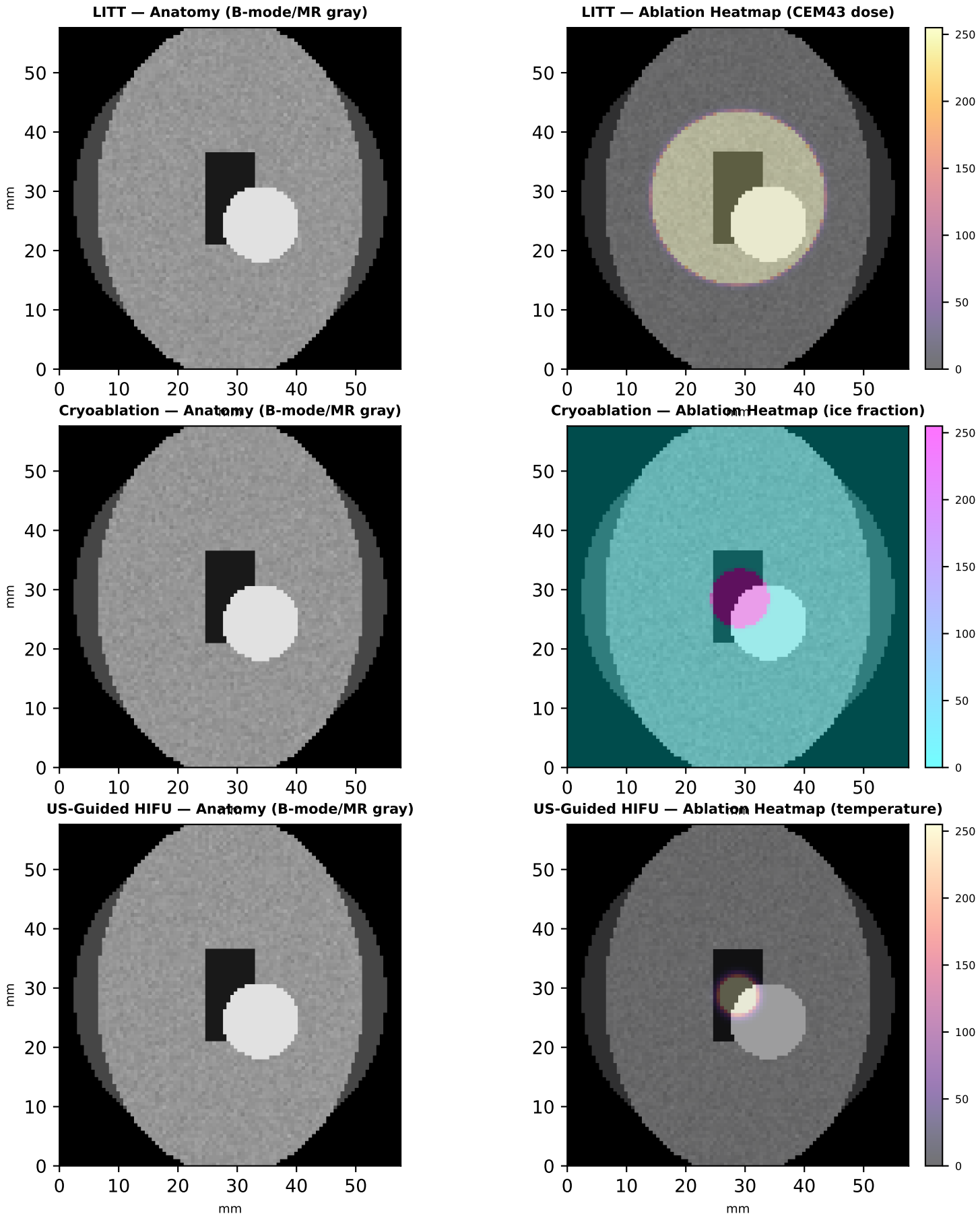
$$F(\phi, m) = \int_0^\phi \tfrac{d\theta}{\sqrt{1 - m \sin^2\theta}}, \quad \text{TRE}_{\text{SLAM}} = \|\mathbf{x}_{\text{FK}} - \mathbf{x}_{\text{target}}\| + 0.01 \, \delta_{\text{CF}}$$

5. Gompertz Regrowth & Weibull Local-Control/Survival

$$\tfrac{dV}{dt} = aV \ln\left(\tfrac{b}{V}\right) \; \Rightarrow \; V(t) = b \exp\left(\ln(V_0/b) \, e^{-at}\right), \quad S(t) = \exp\left(-(t/\lambda)^k\right)$$

Simulation summary (default clinical parameters): LITT peak temperature 100.0 C, CEM43 peak 72001283255745576 min; cryo minimum temperature -180.0 C, Stefan front velocity 205.54 um/s; HIFU peak temperature 95.0 C with TRE 0.080 mm; SLAM converged insertion TRE 4.99 mm.

6. Empirical Grayscale Anatomy & Thermal-Dose Heatmaps



7. SLAM/Stieltjes Trajectory Convergence & Long-Horizon Outcomes

